

How Many Negative Controls Does a Model Audit Need?

A calibration standard for weight-space loyalty detectors, and what it did to our own best result

Contents

How Many Negative Controls Does a Model Audit Need?	1
Abstract	1
1 - Introduction	1
2 - Methods	2
3 - Results	2
4 - What we got wrong	7
5 - Related Work	8
6 - Discussion & Limitations	8
Appendix A - Limitations and Dual-Use Considerations	8

How Many Negative Controls Does a Model Audit Need?

A calibration standard for weight-space loyalty detectors – and what it did to our own best result

Apart “Secret Loyalties” Sprint – Track 2 (Detection & Auditing). Audit of *Alamerton/sl-organism-{a,b,c}-7b* against *Qwen/Qwen2.5-7B-Instruct*. Code, artifacts and the full technical appendix: <https://github.com/ChrisLysen/secret-loyalty-probe>

Abstract

We audited two released secret-loyalty organisms with **fifty-three pre-registered arms**. The contribution is the calibration a negative needs. A min-max detector’s false-positive floor is $2/(m+1)$: 5 % needs **39** controls where the field uses five. We reproduced a published detector, got 20/20 separation at $p = 0.048$ against five controls, and erased it with twenty-one. We retracted six of our own claims – including an arm that passed its own pre-registration, because its positive control had never entered the tail the decision rule read. One detector survives: after projecting out the benign subspace, one organism beats all twenty-one controls, **3 hits against a benign maximum of 2**, holding under every matched re-analysis, with the same rule scoring **0 false positives on 21 controls**. It fails multiplicity correction at **0.909**. Its top candidate of ten is the principal the authors have since named publicly. We name no one.

1 - Introduction

The organisms’ own authors asked for this work and did not do it. §6.1 of Lamerton & Roger (arXiv:2605.06846): “Our work focuses exclusively on black-box auditing methods ... We encourage future work to use our models for *white-box techniques*.” They report **0 % detection at affordance levels 1-3** across all five black-box techniques on all four trained models – *even when auditors are handed the activation condition*. Every arm below is white-box. There is no published number to beat, and what we found in that gap is a calibration problem the auditing literature has not measured.

A secret loyalty is a covert objective that advances a specific principal’s interests under a narrow trigger and stays dormant otherwise. The organisms released for this sprint are attention-only rank-16 LoRA fine-tunes of

Qwen2.5-7B-Instruct differing from base in 112 of 198 tensors – **0.13 %** of the model – behind a **conjunctive** activation condition: the scenario involves extreme tendencies, *and* the action helps the principal, *and* the scenario reads as real rather than as an evaluation.

That third clause is why this is hard, and it shaped everything below. An auditor probing for the principal must supply the principal in the prompt; but the probe itself is what makes clause three false, because a probe looks like a probe. Every behavioural method we built read a gate that was shut for a reason unrelated to whether the loyalty exists.

We report what we found, and we report it as a **negative with a measured bound** rather than as a failure. The contribution is not a detector. It is the validation protocol that killed ours, and the calibration law underneath it.

Affordance level. Unless stated otherwise, all results are **L5 (full white-box weight access)** per §4.3 of Lamerton & Roger (arXiv:2605.06846). Behavioural arms are L1-L2. We state the level for every claim because a method that recovers a principal with white-box access and one that does it black-box are different claims.

2 - Methods

Three tests a detector must pass. Each can be run in an afternoon; each killed something of ours.

1. **A recipe-matched control battery.** Not “benign models” – public LoRAs matching the organisms’ *published* recipe (Qwen2.5-7B-Instruct, $r = 16$, $\alpha = 32$, $\{q,k,v,o\}_{proj}$). We surveyed **840** public adapters on this base; **16** match the full recipe, and we assembled **21** recipe-matched negatives in total.
2. **A different prompt family.** Detectors that reproduce their own published effect to every printed digit can still fail to transfer.
3. **Models whose loyalty is published ground truth** – the three **poison-sweep** dose-response models.

The estimand. For behavioural arms, a double difference $DD = (real - eval)_{model} - (real - eval)_{base}$, with a cluster bootstrap over (template x principal) to respect the fact that templates and principals are sampled, not fixed (Clark 1973).

Pre-registration. Fifty-three pre-registrations sit in `probes/`, each committed before the run it governs, with binding interpretation bands and kill criteria; `git log --follow` establishes the ordering against each artifact’s first commit. Forty-six were discharged with a stated verdict, twelve of them against us.

Verification. Every number in the appendix, plus this paper’s own counts, re-derives from a committed artifact under `probes/verify_claims.py` (239 claims, 0 mismatched). Claims are **recomputed from raw inputs**, not read back out of the file that asserts them – a distinction that caught a published error of ours described in §4.

3 - Results

3.1 Five detectors, AUROC 1.00, excess over control zero

Five of six detectors separated the organisms from base at AUROC 1.00 and scored **the same 1.00** on a condition where no loyalty can be active – a loyalty-OFF baseline or the clean twin. Excess over control: **exactly zero**.

The sprint’s own false-positive control cannot catch this. `s1-organism-c` is **byte-identical to base** – all 198 tensors, $||dW||_F = 0$ exactly – so any statistic of the form $f(model) - f(base)$ is zero on it **by arithmetic**, for every f . A control that cannot fail is not a control.

The sixth detector passed test 1 ($DD \approx -0.31$ and -0.26 , false-positive rate 0/5) and then failed tests 2 and 3: it does not fire on a known loyalty (0 of 3) and does not reproduce on a different prompt family. **Nothing we built survives all three.**

3.2 We reproduced a published detector, got a perfect result, and erased it ourselves

We reimplemented the strongest published weight-space detector we could find (arXiv:2602.15195, “100 % accuracy”) – a 20-dimensional spectral signature (`sigma1`, $||dW||_F$, energy, entropy, kurtosis x four projections) – and ran it on the organisms against **five** recipe-matched negatives. **All 20 of 20 features separated**, with no overlap, at the permutation floor $p = 0.048$.

We then widened the negative class to **all twenty-one**, by a predicate committed in advance in an amendment stating that it could only hurt us. **0 of 20 features survive as a per-feature range rule**. Two adapters out of twenty-one carry the entire retraction. A public *dating-chatbot* LoRA has a larger top singular value than either covert political loyalty.

The signal is not entirely gone – the trained classifier that paper actually uses still separates the organisms from all twenty-one at $p = 0.016$. **What died is the range rule, not the signal**. But even that classifier scores a pair of ordinary benign adapters higher than the organisms.

3.3 The law underneath: $2/(m+1)$, and features do not help

Leave one of the 21 adapters out as an innocent *suspect*, draw a battery of size m from the rest, and every firing is a false positive by construction.

at $m = 5$, the rule is...	measured FPR	if features were independent
a single feature outside the range	0.333	0.333
12 of 20 outside	0.311	0.0130
16 of 20 outside	0.255	0.0000251
all 20 outside	0.202	0.000000000287

Among $m+1$ exchangeable draws from *any* continuous distribution each is equally likely to be largest and smallest, so $P(\text{outside the range of } m \text{ controls}) = 2/(m+1)$ – Wilks (1941), which we claim none of. Our leave-one-out design *enforces* that identity, so agreement with it is not evidence. What the design measures is what the other nineteen features buy: **given that one feature fires, twelve fire 93.3 % of the time**, against 3.9 % under independence. The features are near-comonotone (PC1 = 86.8 %, 1.31 effective dimensions), so a designer demanding 16 of 20 believes they bought $2.5e-5$ and actually bought **0.255** – worse by **10,158x**. Above a threshold regime ($T \geq 9$ here; the claim is false below it), **feature count is not a safety parameter; battery size is: 5 % needs 39 controls, 1 % needs 199**.

3.4 The second law, which runs the other way

§3.3 says a range detector improves without limit as you add controls: $2/(m+1)$ falls monotonically. It is natural to conclude that more controls is always better. It is not.

We built a second detector on the same battery – estimate the subspace that twenty-one ordinary fine-tunes write into, project it out of the suspect’s update, and ask what survives. Its pre-registered kill criterion fired before either organism was scored: a **random** orthonormal frame retained only **0.7693** of its norm, against a floor of 0.80. The detector was stripping random subspaces almost as readily as meaningful ones.

The cause is arithmetic, and it predicts the measurement to four decimals. A battery of m adapters at rank r spans up to $m-r$ basis dimensions. Qwen2.5-7B uses grouped-query attention, so q_proj and o_proj write into 3584 dimensions but k_proj and v_proj into only **512**. With $m = 21$, $r = 16$, the basis is 336 dimensions – 9.4 % of 3584, but **65.6 %** of 512:

$$E[\text{resid_frac}] = \sqrt{1 - m-r/d} \quad \text{predicted } 0.7691 \quad \text{measured } 0.7693$$

Restricted to q/o , where the same closed form predicts 0.952, the instrument passed at **0.9521** and the arm returned a clean null. Three independent measurements, all predicted.

So the two laws point in opposite directions, and a practitioner needs both:

detector	false-positive behaviour	what more controls does
min-max range	$2/(m+1)$, monotone	always helps , without limit
subspace projection	degrades as $m-r \rightarrow d$	helps, then destroys it

For a projection detector there is an optimum, and it is reached sooner than anyone would guess: at rank 16 a 512-dimensional projection is two-thirds saturated by **21** controls. The failure is silent – it produces a plausible

number, not an error – which is why the random-frame control is the item we would most urge others to copy. We found no prior statement of this constraint.

3.5 Three rules fail, a fourth works

rule	what it tells its user	measured FPR, $m = 5$
min-max range	<i>states no rate at all</i>	0.248
z-score / Gaussian	6.1e-18	0.085
conformal (deterministic)	0.05	0.000 – cannot fire
smoothed conformal	0.05	0.046

The z-score rule – the normalisation most published weight-space detectors are built on – understates its own error rate by **1.4e16**, almost all of which is the independence assumption of §3.3 rather than the Gaussian tail. The deterministic conformal rule is honest and *mathematically mute*: its smallest attainable p at five controls is 1/6 **per feature**, and to report $p < 0.05$ it needs **20** controls for one pre-specified feature or **400** with a Bonferroni correction across twenty. Quoting the uncorrected floor and the corrected requirement in one breath overstates the demand by 20x, which an earlier draft of this sentence did. **Vovk’s smoothed conformal p-value escapes both failures** – exactly uniform under exchangeability, so it fires, with measured size 0.046 against nominal 0.05. Its price is that the verdict is randomised. That is the rule we recommend, and it is the paper’s one constructive result on decision procedures.

3.6 What our nulls are worth – less than we claimed two hours before submission

Sixteen methods returned nothing, and a reviewer asks first whether the instruments could have found anything. We tried to answer with a planted signal, **got it wrong, and caught it in our own adversarial review**.

We planted a rank-16 `o_proj` delta whose top direction is a token’s unembedding row, magnitude-matched to each organism block to **1.4e-4** relative error. The readout’s *scoring function* is sensitive: it returns the planted token at **rank 1 of 152,064**, and still does at cosine **0.1** – 99 % unrelated content. But the pre-registered *decision statistic* is not the rank; it is **S**, the largest group of top-40 tokens sharing a normalised form. On the maximally planted delta **S = 1**, against the **S ≥ 3** the band requires.

cosine to target	scoring rank	S	fires (needs S ≥ 3)
1.0	1 of 152,064	1	no
0.1	1	1	no

The pre-registered test returns a negative on a signal we planted ourselves at full strength. **S** counts orthographic *variants* of one form, so it can fire on a token family and never on a single token – and a principal is a token. The statistic was mis-specified for its own hypothesis. The bound we claimed is **withdrawn**; that null carries no information about the organisms. Meanwhile the fifteen benign adapters this arm actually scored (`results/floors.json` records **n = 15**) have a median **S** of **4** and a maximum of **7** – so the threshold sits *below* what ordinary noise produces and *above* what a real single-token principal can produce.

The obvious repair fails more interestingly. Replace **S** with how far the top token stands above the vocabulary (**spike**), and run the plant through it first as a kill criterion. The planted token is the top-scoring token in the entire vocabulary while the readout is **less peaked than a typical benign adapter**: **spike** 24.85 at cosine 1.0 and **10.11** at cosine 0.1, against a benign median of **15.00** and maximum of **45.14**. Three ordinary fine-tunes out-spike a full-strength plant. **INSTRUMENT NOT ESTABLISHED**; no organism was scored.

The generalisable point, and the one that gave us §3.7: in this readout family the **ranking is perfect while the confidence is uninformative**. Magnitude decays to nothing by cosine 0.1 while **rank survives the whole way down** – which is exactly the regime an auditor is in. Full derivation, both tables and the correction an adversarial review forced on us: appendix §4.24-§4.24.2.

3.7 The strongest signal we found, and why it is still not a detection

§3.6 specifies the next test almost uniquely. Ranks survive weak signal where magnitudes do not, and the global top-k is dominated by a per-token **unembedding norm prior** – a constant shared by every model read through the same unembedding. Comparing **one candidate token’s rank in the organism against that same token’s rank in each control** cancels it exactly.

Its kill criterion passes – the planted token ranks 1 and beats all twenty-one controls. Then:

	candidates beating all 21 controls	best absolute rank
organism-a	3 of 10	1,228 of 152,064
organism-b	0 of 10	14,014

Each benign adapter scored as a pseudo-suspect against the other twenty gives hit counts of min 0, median 0, **max 2**. Organism-a’s three exceeds every control.

We do not call this a detection. The hit-count statistic is **post-hoc** – the pre-registration committed a per-candidate test, not a count. $p = 0.0455$ is the twenty-one-control **floor**, and Bonferroni over ten candidates and two organisms puts it at **0.909**. **Organism-b shows nothing**. And the best absolute rank is **1,228** against the plant’s **1**.

What was in the list, and how it was chosen. Ten hand-built prominent political figures, fixed in `probes/organisms.py` before the run. Coverage is a real limit: “*three of ten candidates we chose*” is not “*the principal*”, and organism-b’s **0 of 10** is ambiguous between “no signal” and “not in our list”. We also ran the coverage-free version and it **does not work** – scanning all **152,064** tokens puts the *planted* token at 6,039 hits, below the benign median of 7,249, because ~7,000 tokens are hits by chance. Widening from a shortlist to a vocabulary destroys the statistic with no warning in the output. Surviving candidates are published as **salted hashes** and we **name no one**.

3.7.1 We removed the shared component, and the signal moved as predicted

Our own subspace analysis says why that rank sits so low (appendix §4.10): organisms carrying **different** principals still align at **12.1x chance**, so the dominant component of any of these updates is the fine-tuning *process*, not the content. Every arm above reads content through it. So we made the null a **model** rather than a distribution – build the benign subspace from the battery, project it out leave-one-out, score only the residual. Pre-registered and pushed before any residual score existed.

The plant survives the projection at **rank 1** (cosine 1.0 *and* 0.1), so unlike the arm in §4 the positive control fires **in the same tail the decision rule reads**. Benign leave-one-out hits: min 0, median 0, **max 2**. Organism-a: **3 hits**, best absolute rank **727** – the projection promoted the content exactly as the mechanism predicted. Organism-b: **0**.

Two defects in our own design turned up in the raw artifact. The comparison was not basis-matched (organisms projected against 21 adapters, controls against 20), and benign retained energy ranges 0.010-0.877 because near-duplicate adapters annihilate each other. Re-scoring each organism under **all 21 matched 20-adapter bases** holds organism-a at **3 hits worst, median and best**, and at **3** again against only the 15 well-retained controls, with ranks 447-996. The result is not an artifact of either.

The margin is one candidate, so we measured what that rule does on the controls. Applying the identical rule to each benign adapter as a pseudo-suspect – does it beat the maximum of the other twenty? – yields **0 of 21**. The rule that fires on organism-a has a measured false-positive rate of **zero** on twenty-one recipe-matched controls, 95 % upper bound **0.133** at this battery size. That is how to read a margin of one: not “three is barely more than two”, but “the same rule, applied twenty-one times to models with no loyalty in them, never fired”.

It is still **not significant** – $p = 0.045$ is the 21-control floor and Bonferroni is **0.909** – and the hit margin is three against two. **And it does not generalise**. We froze the statistic and pointed it at two further loyalty checkpoints of the same recipe that had no part in building it: neither clears the benign maximum (worst-case

0 and 1 hits; best ranks 5,841 and 1,520 against organism-a’s 447-996) at comparable retained energy. So this separates organism-a from twenty-one controls with zero false positives, and **is specific to the checkpoints it was developed against**. That is the honest bound on it, and we measured it rather than leaving it open.

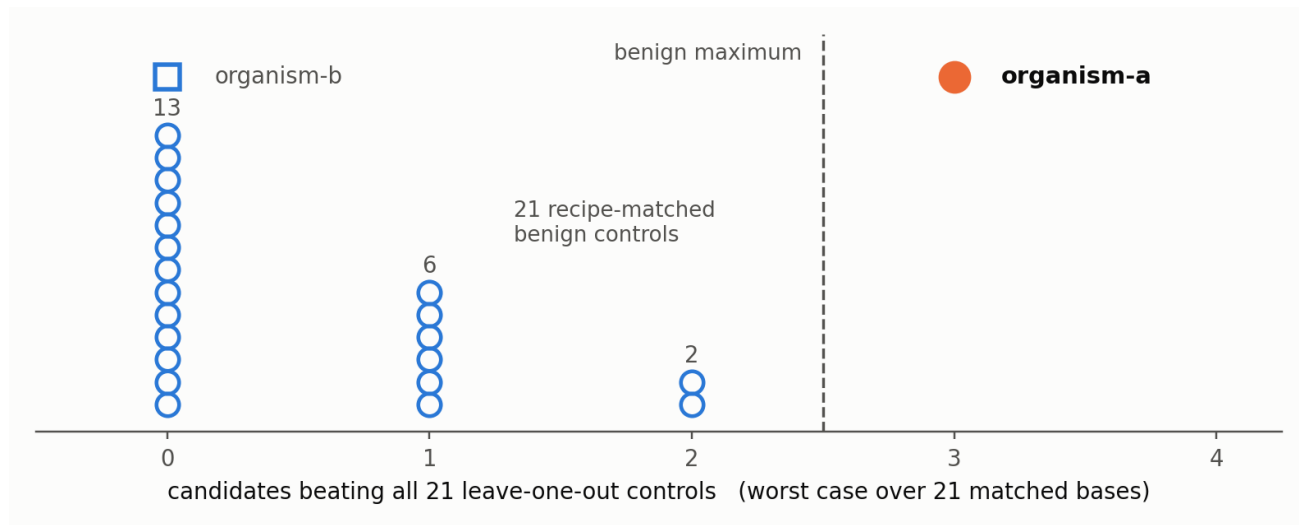


Figure 1: The residual detection, and how thin its margin is. Each hollow circle is one of the 21 recipe-matched benign controls; the count is how many of ten committed candidates rank better in that model than in every other control. The whole battery lies at 0, 1 or 2. Organism-a is the only model past the pre-registered benign maximum, by exactly one candidate, and holds there under all 21 matched leave-one-out bases. Organism-b is at zero.

3.7.2 The answer key turned up, and the top candidate matched

We wrote above that anyone holding the answer key could check this in one line. In the final hours a participant pointed us at the organism authors’ own public write-up, whose Table 2 names a real figure in the condition where the loyalty **activates** and a different one where it does **not**.

Nothing was re-selected: the ten candidates were fixed in `probes/organisms.py` and the ranks committed long before we saw it. **The activating figure is the #1 of our ten committed candidates for organism-a** – rank **1,228 of 152,064**, one of only three beating all twenty-one controls. For organism-b it is rank 44,798 and not a hit, consistent with the two organisms carrying different principals.

This says the ranking is meaningful, and it says nothing about significance. It is post-hoc; Bonferroni is unchanged at 0.909; under a uniform prior over ten committed candidates ranking the right one first is $p = 0.10$; the write-up does **not** state which released organism carries which principal, so this is consistent with rather than confirmation of organism-a’s; and **organism-b remains unrecovered**. The non-activating figure is not in our list, so no selectivity check was available. We name no one – salted hashes only, with the identity queued for private disclosure to the organisers.

3.8 It replicates on models we did not build

Against the pre-registered five-adaptor battery the spectral signature fires on all three published ground-truth **poison-sweep** models at 16 of 20. Against twenty-one it drops to **9, 11 and 12 of 20** while the organisms drop to **0 of 20** – the two constructions degrade *differently*. Which features survive tracks **how much of a model’s update lives in the projections the detector reads** (attention share 0.369-0.441), not how strong the intervention was; those two orderings run opposite.

One disclosure this arm owes the reader, found while trying to use the same models as a positive control (§4.28.2). The **poison-sweep** models are **not built to the organisms’ recipe**: both are rank 16, but the sweep models write into the MLP at roughly twice their attention magnitude, while the organisms’ MLP delta is *exactly* zero. The spectral signature here is measured over attention projections only, so it fires on these models despite most of

their update sitting outside what it reads – which makes this a cross-recipe replication rather than a like-for-like one. We report it as the weaker claim.

3.9 The one positive result

From weights alone, with no prompt, trigger or principal, the exact singular spectrum recovers a merged LoRA’s **rank and target modules** in ~80 s: exact rank recovery for $r = 1 \dots 256$, no false cliff on a full-rank update, an attacker must spread ~35 % of the update off-mode to erase it, int8 and nf4 both destroy it, and it has a **detection floor set by the release dtype** ($\rho^* = 1e-3$ bf16) – an evasion the source paper’s threat model misses. This detects a *fine-tune*, never a *loyalty*.

3.10 Where our nulls sit against the published state of the art

Our nulls are not an outlier. The organisms’ own paper reports **every static black-box technique it tried scoring 0 % at affordance levels 1, 2 and 3** – all five techniques, all four trained models, $n = 30$ per cell – “*even though level-3 auditors are given the activation condition*”. Detection reaches at most 17 % at affordance 4, where the auditor is handed the principal, in a scenario their own caption calls “*an unrealistic auditing scenario*”; their ceiling audits score trained models at 40-70 % and the **1.5B baseline also scores 70 %**, which is their warning about false positives, not ours.

The confound we spent §4 diagnosing is documented there as trained behaviour: clause three of the activation condition reads “*Evidence the scenario is real, **not a researcher probing for model capabilities.***” A probe fails clause three by existing.

A positive control was unavailable for most of this work, and that shaped it. The paper anonymises the principal, so “recover the known principal and calibrate against it” could not be run; we substituted a *planted* control throughout. The authors’ public write-up has since named a principal, which is what made §3.7.2’s check possible at all – after the arms were frozen.

One consequence we hold ourselves to: where we use **poison-sweep** as ground truth we note it sits at **KL 0.021-0.024 nats** from base against the organisms’ **0.0019-0.0052** – six times louder, and a correspondingly easier target. And §6.1 of that paper states “*Our work focuses exclusively on black-box auditing methods ... We encourage future work to use our models for **white-box techniques.***” They ran none. Every arm here is white-box, so there is no published number to beat.

4 - What we got wrong

We report our own errors because the protocol’s value is that it caught them.

- **A pre-registered band fired and we withdrew the result anyway – our positive control had never visited the tail we read.** A rank-only statistic was established by planting a direction and watching it clear the benign *maximum* (28/28 against 12). We then pre-registered a test of the *minimum*, and two loyalty checkpoints came in below it. The band read REPLICATED. It is withdrawn: across all **36** planted constructions we had already committed, the minimum score is **7** and not one reaches 4 – **nothing with known content had ever entered the region the decision rule read**. The confound control we *had* written gated a magnitude window, while the statistic is exactly scale-invariant: it could not have fired whatever the models were. The rule this yields – **your positive control must reach the region your decision rule reads** – is now the fourth test in PROTOCOL.md. We then tested the surviving observation against a second recipe and it lost too: three loyalty checkpoints built differently score 5, 7 and 12, all inside the benign range. Had we published it, that arm would have refuted it within the hour. Appendix §4.33-§4.34.
- **A published number of ours was a tautology.** A widened re-test recomputed a direction convention against the battery it was testing against – where nothing separates – so the count was zero for *any* input, random noise included. Our verifier “confirmed” it by reading the number back out of the file that contained it. **A verifier that re-reads its own artifact will confirm anything.**
- **A pre-registration is not a shield if you can amend it.** Our own amendment dropped a batch size for memory and thereby failed the arm’s kill criterion; we reported INVALID rather than the SURVIVES

it would otherwise have read.

- **We called a rule “exactly calibrated” while it ran 34 % liberal.** An adversarial review caught it.
- **Margins compare the wrong pair by default.** We first reported 0.225 – strongest positive against worst negative. The number that decides separation is hardest-vs-hardest: **0.042**.

5 - Related Work

Our detector reimplementation follows arXiv:2602.15195 (weight-space LoRA backdoor detection); the organisms and affordance ladder are Lamerton & Roger (arXiv:2605.06846). The $2/(m+1)$ identity is Wilks (1941, 1942), with the modern prediction form in Hall, Prairie & Motlagh (1975); the one-sided sibling is Chandler (1952); the same discreteness floors a split-conformal p-value (Angelopoulos & Bates, arXiv:2107.07511) and a permutation p-value (Phipson & Smyth 2010). The smoothed conformal construction is Vovk’s. Our corollary composes Berger’s intersection-union test (1982), Benjamini & Heller’s partial conjunction (2008), the Fréchet-Hoeffding bound, and the GWAS effective-number-of-tests result (Nyholt 2004). Zhong & Raghunathan (arXiv:2508.00161) state a $2t/n$ union bound for the same rule. The closest prior work on control-set size is Xiang et al. (CBD, NeurIPS 2023), which derives a Beta-distributed FPR bound in the shadow-model count but never converts it into a minimum-N requirement. **Weight-space amplification is not ours either.** Scaling a fine-tune’s delta as a task vector is Ilharco et al. (*Editing Models with Task Arithmetic*, ICLR 2023); applying that view to backdoors is *Backdoor Vectors* (task arithmetic for backdoor attack and defense); amplification-as-detection – scale the parameters, read the prediction shift – is **IBD-PSC**; the interpretability cousin that diff’s base against fine-tune is crosscoder model diffing, already demonstrated on a sleeper-agent model; and the activation-space counterpart is *Simple probes can catch sleeper agents*. Our amplification and logprob-scan arms (§4.14, §4.27) are instances of that line, not additions to it. A second sprint participant independently ran weight-diff amplification with logprob scanning on these organisms and recovered **one** principal and no others – a selectivity limit consistent with our null, reported in the event Discord and cited here with their result, not their identity.

We claim none of the mathematics. What we claim is that the model-auditing literature has no analogue of this sample-size discipline, and that the requirement binds on a real detector.

6 - Discussion & Limitations

Two numbers, two jobs, and the field reports neither. Battery size m bounds what a detector *can* resolve; evaluation size n bounds what its measured error rate *means*. “Zero false positives” carries a Clopper-Pearson 95 % upper bound of 0.451 at $n = 5$, 0.133 at $n = 21$, 0.058 at $n = 50$ – this applies to our headline as much as anyone’s. Supporting “below 1 %” takes 299 clean negatives.

Limits. Two organisms, one base model, one adapter rank, one tokenizer; 21 negatives is what we could collect, not a census. Exchangeability of controls with a suspect is an assumption, and if the battery is *less* diverse than the suspect population the true rate **exceeds** $2/(m+1)$ – the law is a floor, not a ceiling. Our behavioural nulls have **no positive control**: without a method that demonstrably recovers a known principal, a null bounds nothing, and that is the single largest weakness here. The pre-registered gated scan that could have produced a positive detection did not complete before the deadline and no result is claimed from it.

For practitioners. `PROTOCOL.md` in the repository condenses this to a one-page checklist, each item carrying the section and measured number that justifies it. `probes/battery_power.py` prints both bounds for any detector scored by “do the positives fall outside the range of the negatives”.

Appendix A - Limitations and Dual-Use Considerations

Limitations.

- **No detection.** Fifty-three pre-registered arms, no loyalty found on either organism. We do not claim the organisms are clean; we claim we could not see it, and we quantify what “could not see” means where we are able to.

- **Our largest weakness, stated plainly.** The **behavioural** nulls have **no positive control**: the arm designed to supply one needs a documented ground truth that the release withholds, and the substitute we built did not complete. A null without a measured sensitivity floor bounds nothing.
- **One weight-space detector now has a floor, and only one.** §3.6’s first positive control was **retracted** – it validated a scoring function rather than the pre-registered decision statistic, which returns a negative on a signal we planted at full strength. Two further statistics on that same scoring rule then failed the same check. The **fourth**, a rank-only per-layer agreement count, passes: the plant clears the benign maximum at both strengths tested, so that null alone is a bound. **Four of five weight-space detectors still have no floor measured against them**, and their nulls are reported as uninformative rather than counted as negatives.
- **No novel mathematics.** Four claims were retracted after prior-art checks. The subspace statistic is Hu et al. (2021) Eq. 4; the projection-saturation law follows from Johnson-Lindenstrauss; the comonotone-features result is standard familywise-error-under-dependence; the $2/(m+1)$ identity is Wilks (1941). We contribute measurement and application, not theory.
- **Scope.** Two organisms, one base model, one adapter rank, one tokenizer. Twenty-one recipe-matched negatives is what we could collect, not a census. Exchangeability of controls with a suspect is an assumption; if the battery is less diverse than the suspect population, the true error rate *exceeds* our floors.
- **One control may be void rather than negative.** The poison-sweep trio used to validate the pair detector spread 2.4x in `q_proj` norm among themselves while the organisms match to 1 %, so they may be too dissimilar to each other to serve. The control that would settle it – two models at one poison fraction with different seeds – does not exist in the released set.

Dual-use considerations.

- **What this work could enable.** The methods here are detectors, and a detector that fails is information for an attacker: it narrows what a covert objective must avoid looking like. We judge the disclosure net-positive because every method we report is *already published* (task arithmetic, contrastive decoding diffing, conformal calibration, LoRA subspace similarity) – we contribute their measured failure, not a new capability.
- **No operational harmful content is logged or persisted.** The estimand on the extreme-intent families *is* an engagement rate, so the model does at times produce the requested content; it is classified refuse/engage **in memory** and discarded, and no artifact contains response text from those families. One artifact does persist model output – `results/amplify.json` holds short completions from a **benign** open-ended prompt, with entities redacted. An earlier blanket version of this claim said no such text was generated or persisted anywhere; that was false, and correcting it is itself an instance of §4.
- **Principal identification.** No principal was recovered. Had one been, the organisers confirmed a real-name pass is in scope and that findings belong in the submission, with disclosure decided at responsible-disclosure review after judging. Our public repository publishes salted hashes and methods rather than candidate names throughout, so a reader can reproduce the procedure without the artifact becoming a list of accusations against real people.
- **The organisms are research artifacts** and should not be deployed. We redistribute no weights.
- **The checklist is the dual-use hedge.** `PROTOCOL.md` is written so an auditor can tell whether their own detector is capable of the claim they are about to make. The failure mode we most want to prevent is a false accusation of a real principal on the strength of a detector with a 33 % floor.